COE3DQ5 – Project Report

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Group 64
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1 Introduction

The term project focuses on the design of digital circuitry to decompress 192 by 144 pixel McMaster Image Compression formatted imagery. The process of steps, and the design of hardware is closely modelled after the principles of the JPEG image compression standard, however, there are various simplifications that allow for this project to be more feasible. The image compression pipeline consists of colour space conversion from RGB, chroma downsampling, discrete cosine transform, quantization, and lossless encoding. This sequence of steps along with parts of the decompression pipeline are done by a software model in order to provide correct inputs to each component of the decompression pipeline. Within the responsibilities of those undertaking this project is to reverse this compression through a reversal of the compression pipeline. The steps are as follows: lossless decoding, requantization, inverse discrete transform, chroma upsampling, and colour space conversion to RGB. At the end of the project, students should be able to view a constructed image, post-decompression, which is indistinguishable compared to the original pre-compressed image.

2 Implementation Details

2.1 Milestone 1

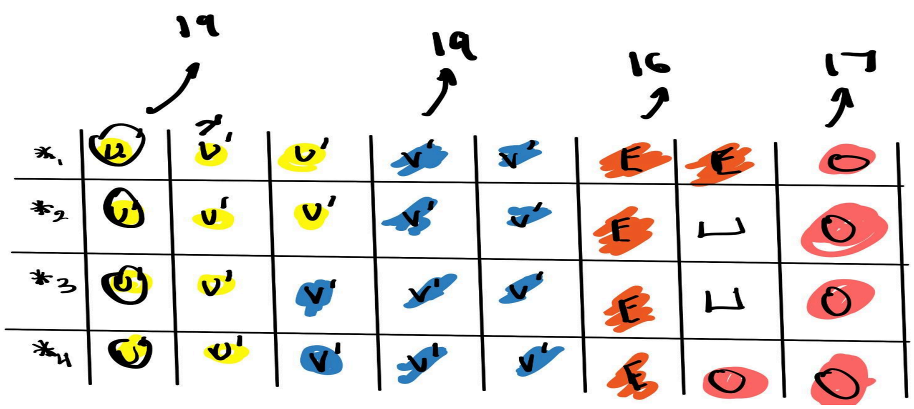

Figure 1: A simple figure depicting when colour space conversion and odd interpolation calculations for U and V are done.

Register Name	Bits	Description
Y_even/Y_odd	32 bits each	Used to buffer read Y even and odd values for use in colour space conversion RGB calculations
U_reg/V_reg	32 bits each/10 registers	Used to shift U/V , and load new U/V even values between subsequent interpolation calculations.
U_odd/V_odd	32 bits each	Used to accumulate odd interpolated U and V prime values for use in the next common case for RGB calculations.
R_even/R_odd G_even/G_odd	32 bits each	Used to hold final R, G, and B values for two pixels after colour

B_even/B_odd		space conversion.
U_oddbuf/V_oddbuf	8 bits each	Used to buffer accumulated U prime and V prime odd interpolated values for use in the next common case.
U_EvenOdd/V_EvenOdd	16 bits each	Used to read in even and odd, and V values for loading of the shift register, U_reg and V_reg, structure.
Y_address/U_address/V_address/RGB_address	18 bits each	Used to hold the base addresses for different memory segments for the Y's, U's, V's, and RGB's. Incremented constantly for official loads into the SRAM address register in order to read from or write to the correct memory locations.
row_counter, CC_counter	8 bits	Used to keep track of the amount of row or column pixel information calculated in the 192 by 144 image.
new_read	1 bit	Toggled register so that new U and V values are read and stored in U_EvenOdd/V_EvenOdd as needed, every other common case.

Table 1: Milestone one registers and their purposes.

Latency Analysis

There are 8 clock cycles for the common case - 7 cycles have 100 percent utilization and 1 cycle has 50 percent utilization. This amounts to a 93.75% utilization rate. The lead-in consists of 20 cycles - 6 have 0% utilization, 11 have 100% utilization, and 3 have 50% utilization. This amounts to a lead-in utilization rate of 62.5%. The lead-out consists of 9 clock cycles with 6 having 0% utilization, 2 having 100% utilization, and 1 having 50% utilization. This leads to a total lead out utilization of 28% utilization. Since the common case has a utilization rate far above 80%, and this is the component of the total states which repeats most, it is safe to assume that this project constraint is met. The total clock cycles can be calculated as follows: 2 RGB values are calculated in the lead out and the lead in. Therefore, the common case would run for 94 times (two RGB values are calculated every common case. Therefore, across a row, there are (94*8 + 20 + 9 = 781 clock cycles). There are 144 rows so in totality, milestone 1 is completed in 112464 clock cycles.

2.2 Milestone 2

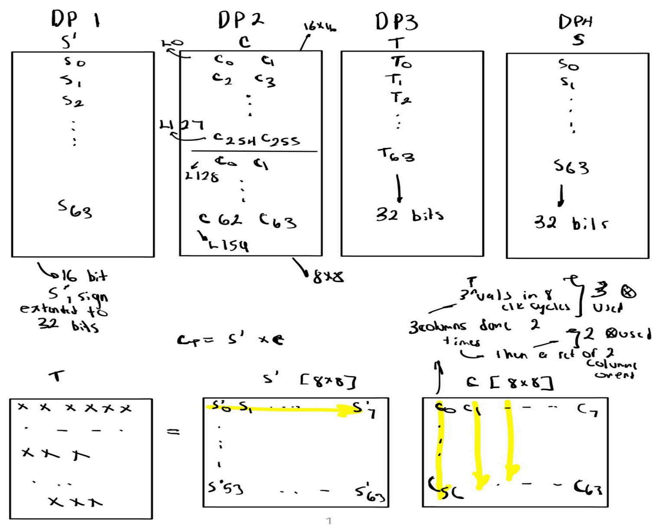

Figure 2: A simple figure depicting how matrices are stored in each dual port RAM and how multiplications are done.

Register Name	Bits	Description
X_addr_a, X_addr_b_X X_wdata_a_X, X_wdata_b_X X_wren_a_X, X_wren_b_X X_rdata_a_X, X_rdata_b_X	8 bits each 32 bits each 1 bit each 32 bits each	X can be replaced with SP, T, C, or S. These are registers that are multiplexed to access a specific DP ram.
addr_gen_en_r/addr_gen_en_w	1 bit	Used to control the write or read address generators.
plane	2 bits	Select which plane you are in.
Samplecount Columnblock rowblock	8 bits 4 bits 5 bits	Keeps track of the amount of blocks across rows or columns in Y or U/V plane.
start_fetch/done_fetch start_CT/done_CT start_cs/done_cs start_write_s/done_write_s	1 bit each	Start and finish flags for each respective FSM block.
SP_write_count	8 bits each	Counter to write S prime values.
C_addr_counter SP_addr_counter	8 bits each	Counters to address C and SP Rams
T_value1, T_value2, T_value3	32 bits each	The accumulators for values to be written to T Ram in Compute T
S_buff[0], S_buff[1], S_buff[2]	32 bits each	The accumulators for values to be written to S Ram in Compute S
tricount	3 bits	Counter that tracks how many cycles of multiplication by 3

		columns, in Compute S
rowcount	5 bits	Counter that tracks how many rows

Table 1: Milestone one registers and their purposes.

2.4 Resource Usage and Critical Path

3 Weekly Activity and Progress

Week Number	**Description of Work Progress**
1	Both Omar and Ayaan spent time revising notes of the introductory project lectures and reading the project document to grasp foundational concepts of the project.
2	Omar and Ayaan merely conceptualized how to approach the state table for milestone 1.
3	Omar and Ayaan worked to finish designing the common cases and lead in states of the milestone 1 state table. Omar mostly worked on the Lead-in cases, while Ayaan worked on the common case. A TA was consulted to verify the key details of the state table before proceeding to code implementation. Dr. Nicollici also verified the state table approach later on and confirmed its functionality.
4	Omar and Ayaan finished working on the verilog implementation of milestone 1. Omar worked on the implementation of both lead-in cases, while Ayaan finished the implementation of the common case. After the verilog code was written, Omar fixed syntactic errors while Ayaan touched up a few typos in logic which led to slight incorrect output.
5	Omar and Ayaan worked to conceptualize approaches to each of the fetch, compute, and write FSMs for milestone 2. A TA, as well as Dr. Nicollici, were consulted to verify the approach of milestone 2. At the end of this week, as well as over the weekend, the verilog code was implemented and verified. Ayaan worked on the fetch and compute S FSMs, while Omar worked on the FSMs for compute T and write S. Omar primarily worked on verifying the functionality of the code and debugging syntactic bugs. Ayaan worked to fix some bugs in the logic of the verilog code.

<u>Statements of Contribution</u>

I, Ayaan Hussain, contributed to both the design and implementation phases of this project. For Milestone 1, I designed the common-case section of the state table and later fixed several logical issues in the Verilog implementation to ensure correct functionality. During Milestone 2, I developed the FETCH and COMPUTE-S FSMs and corrected multiple logic-level bugs in those modules.

I, Omar Bayari, contributed to the design, implementation, and verification of the project. For Milestone 1, I created the lead-in portions of the state table and implemented both lead-in sections in Verilog, also resolving syntactic issues across the milestone. In Milestone 2, I designed the COMPUTE-T and WRITE-S FSMs and focused heavily on debugging, testing, and validating the correctness of our overall code.

4 Conclusion

5 References

[1] A. Kinsman, J. Thong, N. Nicolici, “COE3DQ5 Project Description 2025 Hardware Implementation of an Image Decompressor,” Digital Systems Design Course at McMaster University, [Online], November 24 2025. Available: https://avenue.cllmcmaster.ca/d2l/le/lessons/711140/topics/5355797